Class 4

Nuclear & HEP Astrophysics

A First Exposure for Particle Physicists

Duration: 3 months · 13 weeks · ~5–10 hrs/week
Part of the 2026–2027 Self-Study Plan · RHI → HEP Transition

Deliberately broad tour of the astrophysics most relevant to nuclear and HEP. Stellar nucleosynthesis, compact objects, cosmic rays, neutrinos, dark matter, and multimessenger astronomy.

Course Overview

This class is explicitly labeled “first exposure.” The goal is literacy and motivation across the astrophysics most connected to the physics you and your HEP colleagues care about: stellar nucleosynthesis, compact objects, cosmic rays, astrophysical neutrinos, dark matter, and the pieces of cosmology that touch particle physics.

Do not aim for mastery. Aim to: (a) be able to follow an astrophysics colloquium and ask sensible questions, (b) understand how nuclear/HEP techniques are used in astrophysics, and (c) identify one or two topics you might follow more deeply later.

For an RHI physicist, there are obvious bridges: lattice QCD equations of state for neutron-star interiors, heavy-ion-style simulations applied to neutron-star mergers, and hypernuclear physics. Make note of these connections as you go; they may shape your longer-term HEP direction.

Learning Goals

- Describe stellar structure and the major nuclear reaction networks (pp, CNO, s-, r-, rp-process).
- Explain compact objects (WD, NS, BH) at the level of mass-radius relations and equation-of-state constraints.
- Follow a cosmic-ray or astroparticle-neutrino result paper.
- Describe the standard cosmological model at the level of its particle content.
- Identify specific bridges between your nuclear/HEP work and astrophysics.

Prerequisites & Background Check

Minimal; undergraduate-level thermodynamics and QM are enough for most chapters. Occasional calls on GR intuition can be handled with Hartle.

- Basic stat mech and thermodynamics.
- Familiarity with nuclear physics concepts (binding energy, cross sections, decay chains).
- A rough mental picture of GR (Schwarzschild geometry, event horizons).

Textbooks & Core Resources

Primary Textbooks

- **Iliadis**, *Nuclear Physics of Stars* — the modern nuclear-astrophysics text; spine of Weeks 3–7.
- **Perkins**, *Particle Astrophysics* — accessible, broad; spine of Weeks 8–12.
- **Carroll & Ostlie**, *An Introduction to Modern Astrophysics* — reference for Weeks 1–2 and compact objects.

Supplementary / Reference

- **Rolfs & Rodney**, *Cauldrons in the Cosmos* — classic nuclear-astrophysics companion.
- **Gaisser, Engel & Resconi**, *Cosmic Rays and Particle Physics* — cosmic-ray reference.
- **Shapiro & Teukolsky**, *Black Holes, White Dwarfs, and Neutron Stars* — the compact-objects standard.
- **Kolb & Turner**, *The Early Universe* — particle cosmology classic.
- **Dodelson & Schmidt**, *Modern Cosmology*, 2nd ed.

Online Lectures & Notes

- JINA-CEE webinars and lecture series (joint nuclear astrophysics).
- NASA ADS — practice actually searching papers there.
- NNDC databases (nndc.bnl.gov) — cross sections, reaction rates.
- IceCube, Pierre Auger, LIGO/Virgo, and KM3NeT public summary pages.
- Living Reviews in Relativity & in Solar Physics.
- Annual Reviews of Nuclear and Particle Science — each year has at least one astro-relevant review.

Key Papers & Reviews

- Burbidge, Burbidge, Fowler & Hoyle (B^2FH), *Synthesis of the Elements in Stars* (1957) — read intro and sections on your favorite process.
- Bahcall et al., solar neutrino review papers.
- IceCube Collaboration, diffuse astrophysical neutrino flux papers.
- LIGO+Virgo, GW170817 multimessenger paper (2017).
- Kasen et al., kilonova nucleosynthesis predictions.
- Planck 2018 cosmological parameter paper.
- XENONnT, LZ, PandaX recent dark-matter limit papers.

13-Week Schedule

Broad and fast. Skim rather than drill. The idea is to generate a map of the landscape, not to master a region.

Week	Topic	Readings / Activities	Deliverable
1	Stellar structure essentials	C&O Chs. 10-11; Iliadis Ch. 1	Derive Lane-Emden polytropes
2	Stellar evolution overview	C&O Chs. 13-15	HR diagram annotated
3	Nuclear reactions in stars I	Iliadis Chs. 2-3; Rolfs-Rodney supplement	Compute pp-chain rate at T_c
4	Nuclear reactions in stars II	Iliadis Chs. 4-5	CNO cycle flow diagram
5	s-process, r-process	Iliadis Ch. 5; review papers	Notes on r-process nucleosynthesis sites
6	Supernovae	C&O Ch. 15; Bethe reviews	Summary of neutrino-driven mechanism
7	Compact objects I: WDs, NSs	Shapiro-Teukolsky Chs. 2-3, 8-9	Chandrasekhar limit derivation
8	Compact objects II: BHs, mergers	Shapiro-Teukolsky Ch. 12; GW170817 paper	Notes on kilonova / r-process connection
9	Cosmic rays	Perkins Chs. 6-7; Gaisser excerpts	Sketch the cosmic-ray spectrum annotate
10	Astrophysical neutrinos	Perkins Ch. 9; IceCube astrophysical flux papers	IceCube sensitivity parametrics
11	Dark matter	Perkins Ch. 8; recent WIMP & axion reviews	Notes on direct-detection sensitivity
12	Cosmology essentials	Dodelson-Schmidt Chs. 1-2; Planck 2018 paper	Boltzmann equation for freeze-out sketch
13	Multimessenger synthesis; capstone	GW170817 companion papers; recent NS-merger review	Capstone writeup

Pacing assumes ~5–10 hrs/week on this course while running in parallel with one other class. If time is tighter, cut problem sets first; if time is looser, add problems and deeper paper reading.

Problem Sets & Exercises

Exercises are lighter than in the core classes. Pick one or two per week.

- Iliadis end-of-chapter problems in Chs. 2–5; focus on those with astrophysical numerics.
- Compute the nuclear burning lifetimes for a 1 $M_{\odot}$ star on the main sequence.
- Derive the Chandrasekhar mass from scratch; explore why it's independent of composition.
- Estimate the neutrino flux on Earth from a Galactic core-collapse supernova; compare to Super-K sensitivity.
- Plot the cosmic-ray spectrum from published compilations on a log-log scale; annotate the knee, ankle, GZK features.
- Derive the WIMP thermal relic abundance from the Boltzmann equation (Kolb-Turner style).

Capstone Project

~6 pages, focused. Choose a topic where nuclear/HEP techniques meet astrophysics, and write the review you wish existed.

- Pick a topic. Candidates: kilonova r-process nucleosynthesis; neutron-star equations of state from lattice QCD; sterile neutrinos in cosmology; axion dark matter search techniques; multimessenger constraints on NS merger remnants.
- Motivate why it matters for particle physics.
- Describe the relevant astro observations and the theoretical machinery.
- Identify 1–2 open questions and the measurements that would address them.
- Conclude with: “How would my RHI/HEP skills contribute to this?” — one explicit paragraph.

Milestones & Self-Assessment

At each checkpoint, you should be able to do the listed tasks without looking them up. If you can't, go back rather than pushing forward.

- **End of Week 4:** You can sketch, without notes, the H and He burning reaction networks and explain why the CNO cycle dominates in massive stars.
- **End of Week 7:** You can explain the TOV equation, why neutron stars have a maximum mass, and why neutron-star mergers are the r-process smoking gun.
- **End of Week 10:** You can describe cosmic-ray acceleration mechanisms (diffusive shock acceleration) and the IceCube high-energy neutrino flux.
- **End of Week 13:** You have a list of concrete research directions you could imagine working on, and you can follow an astro colloquium without Googling every third term.

End of class plan. Iterate after the first two weeks — what was over-ambitious gets cut; what felt thin gets deepened.